Dr. A. Veera Ravi

Professor

Contact

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ResearchGate

Academic Qualifications:

B. Sc., M. Sc., M. Phil., Ph. D.,

Teaching Experience:

25 Years

Research Experience:

25 Years

Additional Responsibilities

- Coordinator of Theme Based Research Project under RUSA 2.0 (Alagappa University)
- Director of Collaborative Programmes (Alagappa University)
- Coordinator of Institutions Innovation Council (Alagappa University)
- Former Dean i/c – College Development Council (Alagappa University)
- Former Director – Curriculum Development Cell (Alagappa University)
- Former Special Officer – Planning and Development (Alagappa University)
- RUSA Deputy Coordinator
- Swachh Bharat and Swasth Bharath Program Officer
- Intellectual Property Cell member
- Start-up cell Coordinator
- Members of advisory committee ALUTES-2019
- Administrative and Technical advisory member for USIC
- Swachhata Ranking of higher education 2019 member

Areas of Research

- **Bacterial Communication System– Quorum Sensing**
- **Identification of anti-QS compounds against bacterial pathogens**
- **Probiotics for Aquaculture**
- **Marine Natural Products**
- **Zebrafish Model System for understanding bacterial infections**
- **Nanomaterials for Aquaculture**
- **Photocalytic bacterial deactivation**

Research Supervision / Guidance

Program of Study		Completed	Ongoing
Research	**NPDF**	02	-
	Ph.D.	10	04
	M.Phil.	04	-
Project	**PG**	50	04
	UG / Others	12	-

Publications

International		National		Others
Journals	**Conferences**	**Journals**	**Conferences**	**Books / Chapters / Monographs / Manuals**
95	90	00	24	10

Cumulative Impact Factor (as per JCR) : 422.501 (Avg. I.F. 4.44)
h-index : 37
i10 index : 73
Total Citations : 4060

Funded Research Projects

Completed Projects

1. Principal Investigator in UGC sponsored project on "Selection and evaluation of potential Probiotics for the commercial production of Mud Crab seeds (*Scylla serrata*, Portunidae)" from 01.04.2007 to 31.03.2008. Sanctioned amount **Rs. 1.00 lakh.**

2. Co-Investigator in DBT sponsored project on "Evaluation of bacterial diversity associated with coral *Acropora digitifera* by 16S rRNA gene sequences for antiviral activity against the aquaculture pathogen blotched Snakehead Virus", from 22.09.2006 to 21.09.2009. Sanctioned amount **Rs. 36.315 lakhs.**

3. Principal Investigator in UGC sponsored major research project on "Quorum quenching mediated attenuation of virulence genes expression of pathogenic *Vibrio harveyi* infection in *Penaeus*

monodon" for the period of three years w.e.f. 01.02.2009 to 31.01.2012 with sanctioned amount of **Rs. 12.34 lakhs**.

4. Co-Principal Investigator in DBT sponsored major research project on "Bioprospecting marine microbial wealth through metagenomics" for the period of three years from 13.09.2010 to 12.09.2013 with sanctioned amount of **Rs. 49. 36 lakhs**.

5. Principal Investigator in DBT sponsored major research project on "Antipathogenic potential of marine cyanobacteria in preventing quorum sensing dependent bacterial infections among aquaculture organisms" for the period of three years from 29.09.2012 to 28.03.2016 with sanctioned amount of **Rs. 55.09 lakhs**.

6. Principal Investigator in Tablets (India) Limited sponsored research project on "Studies on effective control of EMS in the culture of *Litopenaeus vannamei*" for the period of two years from 01.07.2015 to 30.4.2018 with sanctioned amount of **Rs. 22.26 lakhs**.

7. Co-Principal Investigator in Committee for Safety Research Programme (CSRP), Atomic Energy Regulatory Board (AERB), Govt. of India, Mumbai sponsored major research project on "Studies on microbial diversity and ecology in the vicinity of a coastal nuclear power plant in relation to water quality and nutrients" for the period of three years from 21.04.2015 to 20.04.2018 with sanctioned amount of **Rs. 32.22 lakhs**.

8. Principal Consultant in Evolva Biotech Private Limited sponsored research project on "Anti-infective and antipathogenic efficacy of resveratrol against quorum sensing mediated virulence and biofilm formation of aquatic pathogens: A promising alternative strategy to antibiotic use in aquaculture" for the period of three years from 06.07.2017 to 05.07.2020 with sanctioned amount of **Rs. 30.00 lakhs**.

On-going Projects

1. Co-Principal Investigator in DAE-BRNS sponsored research on "Assessment & Monitoring of Biofouling Diversity of Cooling Water System of KKNPP & its control: for the period from 2019-2022 with sanctioned amount of 35.63 lakhs

2. Principal Investigator in RUSA 2.0 TBRP sponsored research on "Translational health Research for Human, Animal and Plant systems" for the period from 2022-2023 with sanctioned amount of 10.94 lakhs

3. Principal Investigator in RUSA Entrepeuneur in residence (EIR) sponsored research on "Construction of a novel membrane nanovesicle prepared using tumor homing bacteria and platelets as a vehicle for carrying therapeutic cargos to target hypoxic tumors" for the period from 2022-2023 with sanctioned amount of 2 lakhs

4. Principal Investigator in RUSA Entrepeuneur in residence (EIR) sponsored research on "Mass cultivation and fabrication of microalgae as an entrepreneurial venture by evaluating its bioactive molecules for the anti-quorum sensing potential against marine pathogens" for the period from 2022-2023 with sanctioned amount of 2 lakhs

5. Principal Investigator in Tamil Nadu State Government project on "Exploration of alternative strategy to antibiotics in bacterial disease management" for the period from 2019-2023 with sanctioned amount of 100 Lakhs

Patents

S. No.	Title	Inventors	Patent Number	Filing Date	Current Status
1.	Phytochemical formulations against early mortality syndrome (EMS)	Ravi AV, Santhakumari S, Durgadevi R, Alexpandi R and Pandian SK	Application No. 201841010325	21.03.2018	Granted
2	Phytochemical formulations against early mortality syndrome (EMS)	Ravi AV, Santhakumari S, Durgadevi R, Alexpandi R and Pandian SK	Application No. 201841010346	21.03.2018	Published
3	An antibacterial composition and implementations thereof	Swetha TK, Pandian SK, Sivasankar C, Balamurugan K, Ravi AV, Bhaskar JP, Venkateswaran K, Deepa M, Das SS	Application No. 201831008480	07.03.2018	Published
4	A composition comprising phytochemicals and applications thereof	Swetha TK, Pandian SK, Sivasankar C, Balamurugan K, Ravi AV, Bhaskar JP, Venkateswaran K, Deepa M, Das SS	Application No. 201831008481	07.03.2018	Published
5	A composition comprising phytochemicals and applications thereof	Swetha TK, Pandian SK, Sivasankar C, Balamurugan K, Ravi AV, Bhaskar JP, Venkateswaran K, Deepa M, Das SS	Application No. 201831008482	07.03.2018	Published
6	Antibacterial composition and uses thereof	Swetha TK, Pandian SK, Sivasankar C, Balamurugan K, Ravi AV, Bhaskar JP, Venkateswaran K, Deepa M, Das SS	Application No. 201831008483	07.03.2018	Published

7	Phytochemical-based antibacterial composition and applications thereof	Abirami G, Ravi AV, Alexpandi R, Roshni PS, Jayakumar R, Karutha Pandian S	Application No. 202241007803	14.02.2022	Filed
8	A composition for aquaculture, process for preparing the composition and implementations thereof	Alexpandi R, Ravi AV, Abirami G, Roshni PS, Jayakumar R, Karutha Pandian S	Application No. 202241007807	14.02.2022	Filed
9	Phytochemical-based antibiofilm composition and applications thereof	Roshni PS, Ravi AV, Alexpandi R, Abirami G, Jayakumar R, Karutha Pandian S	Application No. 202241011677	03.03.2022	Filed

Academic awards and fellowships

2019 : Distinguished Professor Award from DKIR Foundation
1996 : University Gold Metal for the best Ph.D., thesis.
1991-1994 : ICAR Senior Research Fellowship
1986-1988 : UGC's National Merit Scholarship during M.Sc.,

Events organized in leading roles

Number of Seminars / Conferences / Workshops / Events organized: **09**

1. **National Training Programme:**

- **Organizing committee member in Vacation Training Programme on Bioresources for School Children (Funded by National Bioresources Development Board, Department of Biotechnology, Ministry of Science and Technology, Government of India) conducted by the Department of Biotechnology, Alagappa University, Karaikudi, May 03-26, 2006.**

2. **Organizing committee member**

- **The National Seminar on "Biotechnology of Transgenesis: Scientific Progress and Social Issues" organized by Prof & Head Dr. S. Karutha Pandian, Department of Biotechnology, Alagappa University, Karaikudi, 2003.**

- **Co-organizer for the 'Biotechnology Exhibition' conducted by the Department of Biotechnology on Feb 28th of every year as part of National Science Day celebrations for the benefit of the school/college students.**

- **Co-organizer for the "Proceedings of International Conference on Environmental Security for Food and Health" organized by M. D. T. Hindu College, Tirunelveli, K. K. Trust, Chidambaram and EPS Global Medical Development Inc., Canada, February 16-18, 2012.**

- **National Workshop on "Microbial Diversity: DNA to Database" organized by Prof & Head Dr. S. Karutha Pandian, Department of Biotechnology, Alagappa University, Karaikudi, February 23-25, 2012.**

- **UGC and DST sponsored National Seminar on "Role of Microbes in Health, Agriculture and Industry" organized by Prof & Head Dr. S. Karutha Pandian, Department of Biotechnology, Alagappa University, Karaikudi, March 26 & 27, 2012.**

- **UGC, DBT and DST sponsored National Seminar cum workshop on "Microbial Diversity and Infectomics" organized by Prof & Head Dr. S. Karutha Pandian, Department of Biotechnology, Alagappa University, Karaikudi, February 27-March 1, 2013.**

- **One day workshop on Innovations from Academia to Industry on "Efficacy of Phytobiotic Ecomask (TIL, Chennai) against EMS in shrimps" organized by Prof. A. Veera Ravi, Department of biotechnology, Alagappa University, Karaikudi, April-20, 2022.**

- **Coordinator (AUIIC) for the "One day National level Symposium on IPR and Industry – Institutional – MSME collaborations" organized by UBCC, AUIIC of Alagappa University MSME Govt. of India, CIPAM, DPIIT Govt. of India held at Alagappa University, Karaikudi on April 10th, 2023.**

Extension Work / Community Services

(i) Community

1. Offered effective suggestions about water quality management to rural people during the Village Placement Programme in 2005 and 2006 at O. Siruvayal village, Karaikudi.

2. Arranged Siddha and Veterinary camps for the benefit of poor people in O. Siruvayal village during Village Placement Programme in 2005 and 2006.

(ii) Extension works

1. Serving as Programme Officer in Alagappa University National Service Scheme (NSS) from 2005 onwards. Effectively organized various NSS camps at O. Siruvayal, Kandanoor and Sankarapuram villages.

2. Acted as Co-ordinator and Assistant Co-ordinator in Village Placement Programme, organized by Alagappa University for the M.Sc., Biotechnology programme students of 2004 – 2006 and 2005 – 2007 set at O. Siruvayal, 2012-2013 set at Ariyakudi, Karaikudi, 2014 – 2016 set at Nemam, Karaikudi.

3. Organizing Committee member in "Science Exhibition programme for school students" jointly organized by National Service Scheme unit, Alagappa University and Rotary Club, Karaikudi, August 30, 2013.

4. Organizing Committee member in "Blood Donation Camp" jointly organized by National Service Scheme unit, Alagappa University and HDFC Bank, Karaikudi, December 6, 2013.

5. Organizing Committee member in "Science Exhibition programme for school students" jointly organized by National Service Scheme unit, Alagappa University and Rotary Club, Karaikudi, September 1, 2014.

6. Organizing Committee member in "Blood Donation Camp" jointly organized by National Service Scheme unit, Alagappa University and HDFC Bank, Karaikudi, December 5, 2014.

7. Organizing Committee member in "Tree Planting Campaign" organized by National Service Scheme unit, Alagappa University, Karaikudi, December 11, 2015.

8. Organizing Committee member in "7 Days special camp" organized by National Service Scheme unit, Alagappa University, Karaikudi, March 11-17, 2016.

Events Participated (optional)

Conferences / Seminars / Workshops: 114

. Membership in Professional & Academic Bodies

Professional Bodies

1. Life Member: Society of Biological Chemists, India (SBC)
2. Life Member: The Indian Science Congress Association
3. Life Member: Proteomics Society, India (PSI)
4. Life Member: Biotech Research Society, India (BRSI)
5. Life Member: Association of Microbiologists of India (AMI)

Academic Bodies (such as Board of Studies etc.,)

1. M.Sc., Biotechnology, Alagappa University (Till date)
2. M. Sc., Microbiology, Alagappa University (2011-2014)
3. B. Sc., Microbiology (Chairman), Alagappa University (2011-2014)
4. B. Sc., Biotechnology, Alagappa University (2012-2015)
5. M. Phil., Biotechnology, Alagappa University (2013)
6. B. Sc., Advanced Zoology & Animal Biotechnology (2015-2017)
7. B. Sc., Biotechnology, Alagappa University (2015-2017)
8. B.Sc., and M.Sc., Biotechnology, Jamal Mohamed College, Thiruchirappalli (2015-2017)
9. M.Sc., Microbiology, Madurai Kamaraj University, Madurai (2019-Till date).
10. B.Sc., and M.Sc., Biotechnology, Cauvery College, Thiruchirappalli (2019-Till date)
11. B.Sc., and M.Sc., Microbiology, Cauvery College, Thiruchirappalli (2019-Till date)
12. Member- Result Passing Board, Thassim Beevi Abdul Kader College For Women, Kilakarai.

Number of Invited / Special Lectures delivered: **31**

Others

1. Articles published in Newspapers / Magazines : 10 **Book chapters**
2. Products developed : 0**1**
3. No. of PhD Thesis evaluated : **30**
4. No. of PhD Public Viva Voce Examination conducted : **21**
5. Sequences submitted in GenBank: **141**

Recent Publications (selective)

1. Angusamy, A., Balasubramanian, V., Arunmurugan, B., Arunachalam, K., Issac Abraham, S. V. P., Murugesan, S., ... & Arumugam, V. R. (2023). Anti-infective potential of plant-derived quorum sensing inhibitors against multi-drug resistant human and aquatic bacterial pathogens. World Journal of Microbiology and Biotechnology, 39(6), 147. (**IF: 4.253**)

2. Roshni Prithiviraj Swasthikka, Rajaiah Alexpandi, Gurusamy Abirami, Ravindran Durgadevi and Arumugam Veera Ravi*. (2023) Hesperidin methyl chalcone, a citrus flavonoid, inhibits *Aeromonas hydrophilla* infection mediated by quorum sensing. Microbial Pathogenesis. (**IF. 3.848**)

3. Abirami, G., Alexpandi, R., Sudhin, S., Durgadevi, R., Roshni, P. S., Kumar, P., & Ravi, A. V. (2023). Pyrogallol downregulates the expression of virulence-associated proteins in Acinetobacter baumannii and showing anti-infection activity by improving non-specific immune response in zebrafish model. International Journal of Biological Macromolecules, 226, 853-869. (**IF: 8.025**)

4. Alexpandi, R., Abirami, G., Murugesan, B., Durgadevi, R., Swasthikka, R. P., Cai, Y., & Ravi, A. V. (2023). Tocopherol-assisted magnetic Ag-Fe3O4-TiO2 nanocomposite for photocatalytic bacterial-inactivation with elucidation of mechanism and its hazardous level assessment with zebrafish model. Journal of Hazardous Materials, 442, 130044. (**IF: 14.224**)

5. Lakshmi, Selvaraj Alagu, Rajaiah Alexpandi, Raja Mohamed Beema Shafreen, Kannapiran Tamilmuhilan, Adimoolam Srivathsan, Thirupathi Kasthuri, Arumugam Veera Ravi, Sugathan Shiburaj, and Shunmugiah Karutha Pandian (2022). Evaluation of antibiofilm potential of four-domain α-amylase from Streptomyces griseus against exopolysaccharides (EPS) of bacterial pathogens using Danio rerio. Archives of Microbiology. 1-10. **(IF: 2.552)**

6. Alexpandi, R., Abirami, G., Balaji, M., Jayakumar, R., Ponraj, J. G., Cai, Y. & Ravi, A. V. (2022). Sunlight-active phytol-ZnO@ TiO2 nanocomposite for photocatalytic water remediation and bacterial-fouling control in aquaculture: A comprehensive study on safety-level assessment. Water Research, 212, 118081.. **(IF: 11.236)**

7. Alexpandi, Rajaiah, Gurusamy Abirami, Lakkakula Satish, Roshni Prithiviraj Swasthikka, Nataraj Krishnaveni, Rangarajan Jayakumar, Shunmugiah Karutha Pandian, and Arumugam Veera Ravi (2021). Tocopherol and phytol possess anti-quorum sensing mediated anti-infective behavior against *Vibrio campbellii* in aquaculture: An *in vitro* and *in vivo* study. Microbial Pathogenesis 161. **(IF: 3.738)**

8. Manivannan, Arun Chandra, Vinitha Devaraju, Palanivel Velmurugan, Thangavelu Sathiamoorthi, Subpiramaniyam Sivakumar, Suresh Kumar Subbiah, and Arumugam Veera Ravi (2021). "Tumorigenesis and diagnostic practice applied in two oncogenic viruses: Epstein Barr virus and T-cell lymphotropic virus-1—Mini review. Biomedicine & Pharmacotherapy. 142. **(IF: 6.529)**

9. Alexpandi, Rajaiah, Jeyaraj Godfred Ponraj, Roshni Prithiviraj Swasthikka, Gurusamy Abirami, Thennarasu Ragupathi, Rengarajan Jayakumar, and Arumugam Veera Ravi (2021). Anti-QS mediated anti-infection efficacy of probiotic culture-supernatant against *Vibrio campbellii* infection

and the identification of active compounds through *in vitro* and *in silico* analyses. Biocatalysis and Agricultural Biotechnology. 35. **(IF: 3.28)**

10. Velmurugan, P., Kumar, R.V., Sivakumar, S. and Ravi, A.V (2021). Fabrication of blue fluorescent carbon quantum dots using green carbon precursor *Psidium guajava* leaf extract and its application in water treatment. Carbon Letters. 1-11. Jun. **(IF: 0.380)**

11. Velmurugan, P., Venil, C.K., Ravi, A.V. and Dufossé, L (2021). Marine bacteria is the cell factory to produce bioactive pigments: A prospective pigment source in the ocean. Frontiers in Sustainable Production of Bioactive Pigments. Nov.

12. Packiavathy, I.A.S.V., Kannappan, A., Thiyagarajan, S., Srinivasan, R., Jeyapragash, D., Paul, J.B.J., Velmurugan, P. and Ravi, A.V (2021). AHL-Lactonase Producing Psychrobacter sp. From Palk Bay Sediment Mitigates Quorum Sensing-Mediated Virulence Production in Gram Negative Bacterial Pathogens. Frontiers in microbiology. Apr; 12 748. (**IF: 4.235**)

13. Suresh G, Balasubramanian B, Ravichandran N, Ramesh B, Kamyab H, Velmurugan P, Siva GV, Ravi AV (2021). Bioremediation of hexavalent chromium-contaminated wastewater by Bacillus thuringiensis and Staphylococcus capitis isolated from tannery sediment. Biomass Conversion and Biorefinery. Apr; 11(2): 383-91. (**IF: 2.15**)

14. Velu M, Balasubramanian B, Velmurugan P, Kamyab H, **Ravi AV**, Chelliapan S, Lee CT, Palaniyappan J (2021). Fabrication of nanocomposites mediated from aluminium nanoparticles/*Moringa oleifera* gum activated carbon for effective photocatalytic removal of nitrate and phosphate in aqueous solution. Journal of Cleaner Production. Jan 25; 281: 124553. (**IF: 7.246**)

15. Rashiya N, Padmini N, Ajilda AA, Prabakaran P, Durgadevi R, **Ravi AV**, Ghosh S, Sivakumar N, Selvakumar G (2021). Inhibition of biofilm formation and quorum sensing mediated virulence in Pseudomonas aeruginosa by marine sponge symbiont Brevibacterium casei strain Alu 1. Microbial Pathogenesis. Jan 1;150:104693. (**IF: 2.914**)

16. Prasath KG, Alexpandi R, Parasuraman R, Pavithra M, **Ravi AV**, Pandian SK (2021). Anti-inflammatory potential of myristic acid and palmitic acid synergism against systemic candidiasis in *Danio rerio* (Zebrafish). Biomedicine & Pharmacotherapy. Jan 1;133:111043. (**IF: 4.545**).

17. Abirami G, Durgadevi R, Velmurugan P, Ravi AV (2020). Gene expressing analysis indicates the role of Pyrogallol as a novel antibiofilm and antivirulence agent against Acinetobacter baumannii. Archives of Microbiology. Sep 11: 1-0. (**IF: 1.835**).

18. Durgadevi R, Abirami G, Swasthikka RP, Alexpandi R, Pandian SK, **Ravi AV** (2020). Proteomic analysis deciphers the multi-targeting antivirulence activity of tannic acid in modulating the expression of MrpA, FlhD, UreR, HpmA and Nrp system in *Proteus mirabilis*. International Journal of Biological Macromolecules. Sep 29. (**IF: 5.162**).

19. Srinivasan R, Devi KR, Santhakumari S, Kannappan A, Chen X, **Ravi AV**, Lin X (2020). Anti-quorum Sensing and Protective Efficacies of Naringin Against Aeromonas hydrophila Infection in Danio rerio. Frontiers in Microbiology. Dec 3;11:3087. (**IF: 4.235**).

20. Durgadevi R, Kaleeshwari R, Swetha TK, Alexpandi R, Pandian SK and **Ravi AV** (2020). Attenuation of Proteus mirabilis colonization and swarming motility on indwelling urinary catheter by antibiofilm impregnation: an *in vitro* study. Colloids and Surfaces B: Biointerfaces, 111207. (**IF: 4.389**).

21. Venkatramanan M, Sankar Ganesh P, Senthil R, Akshay J, **Ravi AV**, Langeswaran K, Vadivelu J, Nagarajan S, Rajendran K, Shankar EM (2020). Inhibition of Quorum Sensing and Biofilm Formation in *Chromobacterium violaceum* by Fruit Extracts of Passiflora edulis. ACS Omega. Sep 29. (**IF: 2.58**).

22. Pravinkumar M, Ponraj JG, Ravi AV (2020). Efficacy of Evaluation of Gut Probiotics against White Gut and White Feces Disease in Litopenaeus vannamei Shrimp Aquaculture Systems in Two Different Geographical Regions of Andhra Pradesh. International Journal of Marine Science. Sep 18; 10. (**IF: 3.593**).

23. Saikishore R, Velmurugan P, Ranjithkumar D, Latha R, Sathiamoorthi T, Arun A, Ravi AV, Sivakumar S (2020). The circular RNA-miRNA axis: A Special RNA signature regulatory transcriptome as potential biomarker for oral squamous cell carcinoma. Molecular Therapy-Nucleic Acids. Sep 6. (**IF: 7.032**).

24. Elango D, Manikandan V, Jayanthi P, Velmurugan P, Balamuralikrishnan B, **Ravi AV**, Shivakumar MS (2020). Selection and characterization of extracellular enzyme production by an endophytic fungi *Aspergillus sojae* and its bio-efficacy analysis against cotton leaf worm, Spodoptera litura. Current Plant Biology. 23:100153. (**IF: 3.26**).

25. Alexpandi R, Chandu VV. Muralee G, Durgadevi R, Hee-Je Kim, Pandian SK and Ravi AV (2020). Metal sensing-carbon dots loaded TiO2 nanocomposite for photocatalytic bacterial deactivation and control acute-hepatopancreatic necrosis disease (AHPND) in aquaculture. Scientific Reports, 10: 12883. (**IF: 3.998**).

26. Alexpandi R, De Mesquita JF, Pandian SK and **Ravi AV** (2020) Quinolines-Based SARS-CoV-2 3CLpro and RdRp Inhibitors and Spike-RBD-ACE2 Inhibitor for Drug-Repurposing Against COVID-19: An in silico Analysis. Frontiers in Microbiology. 11:1796. (**IF: 4.235**).

27. Venil CK, Velmurugan P, Dufossé L, Devi PR, **Ravi AV (**2020). Fungal Pigments: Potential coloring compounds for wide ranging applications in textile dyeing. Journal of Fungi. Jun;6(2):68. (**IF: 4.621**).

28. Balaji M, Nithya P, Kasirajan K, Alexpandi R, Mayakrishnan A, Palanisamy S, Jegatheeswaran S, Selvam S, **Ravi AV**, Karunakaran M, Cai Y (2020). Fabrication of heteroatom doped NFP-MWCNT and NFB-MWCNT nanocomposite from imidazolium ionic liquid functionalized MWCNT for antibiofilm and wound healing in Wistar rats: Synthesis, characterization, *in-vitro* and *in-vivo* studies. Materials Science and Engineering: C, 110791 (**IF: 5.88**).

29. Kannappan A, Durgadevi R, Srinivasan R, Lagoa R, Packiavathy IASV, Pandian SK and **Ravi AV** (2020). 2-Hydroxy-4-methoxybenzaldehyde from *Hemidesmus indicus* is antagonist to Staphylococcus epidermidis biofilm formation. Biofouling, (Article accepted, in press) (**IF: 2.786**).

30. Princely SX, Puja P, Vinita MN, Devan U, Velangani AJ, Sunita S, Yuvakkumar R, Velmurugan P, **Ravi AV**, Govarthanan M, Kumar P (2020). Anti-proliferative and anti-migratory effects of flower-like bimetallic (Au@ Pt) nanoparticles. Materials Letters, 267, 127491 (**IF: 3.019**).

31. Suresh G, Kokila D, Suresh TC, Kumaran S, Velmurugan P, Vedhanayakisri KA, Sivakumar S, **Ravi AV** (2020). Mycosynthesis of anticancer drug taxol by *Aspergillus oryzae*, an endophyte of Tarenna asiatica, characterization, and its activity against a human lung cancer cell line. Biocatalysis and Agricultural Biotechnology, 24, 101525. (**IF: 4.26**).

32. Abirami, G., Alexpandi, R., Durgadevi, R., Kannappan, A. and Ravi, A.V (2020). Inhibitory effect of morin against *Candida albicans* pathogenicity and virulence factor production: An in vitro and in vivo approaches. Frontiers in microbiology. 11. (**IF: 4.325**).

33. Lee, J.H., Velmurugan, P., Ravi, A.V. and Oh, B.T (2020). Green and hydrothermal assembly of reduced graphene oxide (rGO)-coated ZnO and Fe hybrid nanocomposite for the removal of nitrate and phosphate. Environmental Chemistry and Ecotoxicology. 2,141-149. (**IF: 3.742**).

34. Durgadevi, R., Abirami, G., Alexpandi, R., Nandhini, K., Kumar, P., Prakash, S. and Veera Ravi, A (2019). Explication of the Potential of 2-Hydroxy-4-Methoxybenzaldehyde in Hampering

Uropathogenic Proteus mirabilis Crystalline Biofilm and Virulence. Frontiers in microbiology. 10, 2804 (**IF: 4.325**).

35. Alexpandi R, Prasanth MI, **Ravi AV**, Balamurugan K, Durgadevi R, Srinivasan R, De Mesquita J, Pandian SK (2019). Protective effect of neglected plant *Diplocyclos palmatus* on quorum sensing mediated infection of *Serratia marcescens* and UV-A induced photoaging in model *Caenorhabditis elegans.* Journal of Photochemistry and Photobiology B: Biology. 201, 111637. (**IF: 4.383**).

36. Kannappan A, Santhakumari S, Srinivasan R, Pandian SK, **Ravi AV** (2019). *Hemidesmus indicus*, a traditional medicinal plant, targets the adherence of multidrugresistant pathogens to form biofilms. Biocatalysis and Agricultural Biotechnology, 21, 101338. (**IF: 4.26**).

37. Ganesh, P. S., Krishnamurthy, V., Iswamy, K., Suvaithenamudhan, S., Amuthan, M., Vimali, I., & Shankar, E. M. (2019). Biofilm-associated Agr and Sar quorum sensing systems of methicillin-resistant Staphylococcus aureus are inhibited by fruit extracts of Illicium verum.

38. Kannappan A, Srinivasan R, Nivetha A, Annapoorani A, Pandian SK, **Ravi AV** (2019). Anti-virulence potential of 2-hydroxy-4-methoxybenzaldehyde against methicillin-resistant *Staphylococcus aureus* and its clinical isolates. Applied Microbiology and Biotechnology, 103, 6747-6758. (**IF: 3.340**).

39. Packiavathy, I.A.S., Maruthamuthu, S., Gnanaselvan, G., Manoharan, S., Paul, J.B.J., Annapoorani, A., Kannappan, A. and Ravi, A.V (2019). The control of microbially induced corrosion by methyl eugenol–A dietary phytochemical with quorum sensing inhibitory potential. Bioelectrochemistry. 128, 186-192. (**IF: 3.789**).

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